Software Requirement Specifications

AI-Based University Queries Chatbot

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Summary

The project aims to develop an intelligent question-answering system designed primarily for students and guests of a university or educational institute. This system enables users to type their queries related to university services, academics, administration, and other relevant topics. By leveraging advanced Natural Language Processing (NLP) techniques, the system understands the intent and context of the user's questions to provide accurate and relevant answers.

The core functionality involves processing user queries automatically to retrieve responses from a pre-built knowledge base. In cases where the system cannot find a satisfactory answer, the query is escalated to an administrator for manual intervention, ensuring that all user questions receive attention and appropriate responses. This hybrid approach combines the efficiency of AI-based automation with human oversight, enhancing reliability and user satisfaction.

The project focuses on creating an interactive and user-friendly interface for easy question submission and prompt response display. It aims to reduce the workload on university staff by automating routine inquiries, improve accessibility of information, and offer a scalable solution that can evolve with expanding data and user needs.

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1. Introduction

The Islamia University of Bahawalpur provides a wide range of academic, administrative, and support services to over thousands of students and faculty. The AI-Based University Queries Chatbot aims to automate frequently asked queries, streamline information retrieval, and reduce load on university helpdesks by simulating human-like interactions.

1.1 Purpose

The purpose of this document is to specify the software requirements for the development of an AI-based university queries chatbot. This chatbot system is designed to assist students, faculty, and administrative staff in obtaining quick and accurate responses to queries related to university services, academics, admissions, and general information.

1.2 Scope

This software product will provide an interactive chatbot, powered by OpenAI, accessible via a user-friendly interface developed in Streamlit. The system will operate on a local server and utilize a SQLite database to manage and store user interactions, FAQs, and administrative configurations. It will simulate human-like interactions, enhancing the user experience while reducing the workload on university help desks.

1.3 Product Perspective

The AI-based University Queries Chatbot is designed as an independent web-based application that integrates with the university's existing information systems and databases. It serves as a user-friendly interface to provide quick and automated responses to common student and visitor queries.

This product complements traditional support services by reducing the volume of routine inquiries handled by university staff. It leverages AI and NLP technologies to understand and process natural language questions, ensuring accurate and relevant answers.

The chatbot system is scalable and can be expanded in the future to include additional features such as voice input, multi-language support, and integration with other university platforms like student portals and learning management systems.

1.4 User Characteristics

- **Students:**
  Primary users of the system who will use it to ask questions related to academics, admissions, exams, schedules, and campus facilities. They generally have basic to moderate computer skills and expect quick and accurate responses.

- **Guests/Visitors:**
  External users such as prospective students, parents, or visitors who seek information about the university, its programs, admission criteria, and general inquiries. They might have limited familiarity with the system and require a simple and intuitive interface.
- **Administrators:**
  Internal users responsible for managing the system's knowledge base and handling escalated queries that the AI cannot answer. They have advanced technical knowledge and are capable of reviewing, updating, and improving the system's responses.
- **University Staff:**
  Faculty and administrative staff who might use the system to provide information or monitor common queries. They are generally familiar with university processes and can contribute to ensuring the accuracy of information provided by the system.

1.5 Similar apps and systems/Literature Review

Several existing chatbot systems have influenced the design of this AI-based University Queries Chatbot:

- **IBM Watson Assistant:** A powerful AI chatbot platform used in education for automating student support with customizable workflows.
- **Google Dialogflow:** Provides natural language understanding for building conversational interfaces, widely used in academic settings.
- **Microsoft Azure Bot Service:** Offers scalable chatbot solutions integrated with AI for handling FAQs in universities.
- **QnA Maker:** A quick-to-deploy knowledge-based chatbot service commonly used for automating responses in educational domains.
- **NLP-based University Chatbots:** Recent research emphasizes using NLP and machine learning to improve chatbot accuracy and user experience in universities

1.6 Proposed Technologies

For Front End :

- HTML
- CSS
- JavaScript

For Back End :

- Python (Flask) (For API Server)
- OpenAI GPT API (AI Chatbot functionality)

- MySQL

2. Requirements

The system will allow users to ask questions through a chat interface. Queries will be processed by OpenAI API and responses displayed instantly. Admins can manage FAQs and monitor chat logs.

2.1 Functional Requirements

ID	Requirement	Input	Output	Priority
FR001	User inputs a question	User query	Processed input	High
FR002	NLP processes the question	Parsed query	Intent and entities	High
FR003	Answer fetched from DB	Intent	Answer text	High

2.1.1 Sign up and login :

- ✓ **User Sign-Up:**
  The system should allow a new user (student or guest) to sign up by providing basic information such as name, email, and password. Upon successful submission, the system creates a new user account. This requirement ensures that users can have a personalized experience and access history-based features. *(Priority: Medium)*
- ✓ **User Login:**
  Registered users must be able to log in to the chatbot system using their email and password. After successful authentication, users are granted access to their dashboard or the chatbot interface. This feature is essential for secure access and personalized interactions. *(Priority: Medium)*

2.2 Non-Functional Requirements

ID	Requirement	Input	Output	Priority
NFR001	Response time < 2 seconds	User query	Response within 2s	High
NFR002	99.9% system uptime	System time	High availability	High
NFR003	Multi-device support	Device type	Responsive UI	Medium

NFR004	Data encryption	User data	Encrypted storage	High
NFR005	Multi-language support	Localized UI	Language adaptability	Medium

security :

Security is a critical non-functional requirement for the AI-based University Queries Chatbot. The following measures ensure data protection, secure access, and system integrity:

1. **User Authentication & Access Control**
   - Only registered users and admins can access restricted features.
   - Guest users have limited access to basic functionality.
2. **Data Encryption**
   - All user input and stored responses are encrypted using standard encryption protocols (e.g., AES-256).
   - Communication between frontend and backend is secured with HTTPS.
3. **SQL Injection Prevention**
   - The system uses parameterized queries or ORM to prevent SQL injection attacks.
4. **API Key Protection**
   - OpenAI API keys are stored securely and not exposed in frontend code.
   - Backend ensures secure and authenticated use of external APIs.
5. **Session Management**
   - User sessions are timed out after inactivity.
   - Secure session tokens are used to prevent hijacking.
6. **Admin Privileges**
   - Admin panel is protected with an additional layer of authentication.
   - Only verified admins can update FAQs or respond to escalated queries.
7. **Data Privacy Compliance**
   - The system avoids storing sensitive personal data unnecessarily.
   - It follows basic data protection principles, ensuring user privacy

3. Use Cases and Flow of Processes

This section outlines the flow of processes that are designed according to the system's defined functional requirements. The AI-based University Queries Chatbot is developed to provide automated, real-time assistance primarily to students and prospective applicants. Its main purpose is to efficiently handle and respond to frequently asked questions related to admissions, courses, schedules, campus facilities, and other university-related information.

The interaction begins when a student or user submits a query through the chatbot interface. The chatbot then processes the input using natural language understanding techniques to comprehend

the intent behind the question. Based on this understanding, the system retrieves the most relevant answers from its knowledge base or generates a response using AI models

3.1 Use Case 1

Field	Details
ID	UC002
Name	Ask Question
Description	This use case describes the process of a student asking a query from the AI-based chatbot.
Requirement(s)	FR002
Actor(s)	Student, Guest User
Precondition	Actor must be logged in or access as guest
Postcondition	Actor receives a relevant answer or prompt for clarification
Basic Flow	1. Actor opens chatbot interface 2. Actor types a query 3. System processes the query using NLP 4. System fetches appropriate answer from knowledge base 5. Answer is displayed to the actor
Alternative Flow	1. If no answer found, chatbot prompts for clarification or redirects to admin
Exceptions	1. Invalid query format 2. System error or database not reachable

Explanation :

This use case outlines the interaction between a student or guest user and the AI-based chatbot system. It describes the steps taken when a user wants to ask a question, starting from opening the chatbot interface to receiving a response. The system uses Natural Language Processing (NLP) to understand the user's query and fetch a relevant answer from the knowledge base. If the system cannot find a suitable answer, it prompts for clarification or escalates the query to an admin. This use case ensures users get timely and accurate support through an automated process.

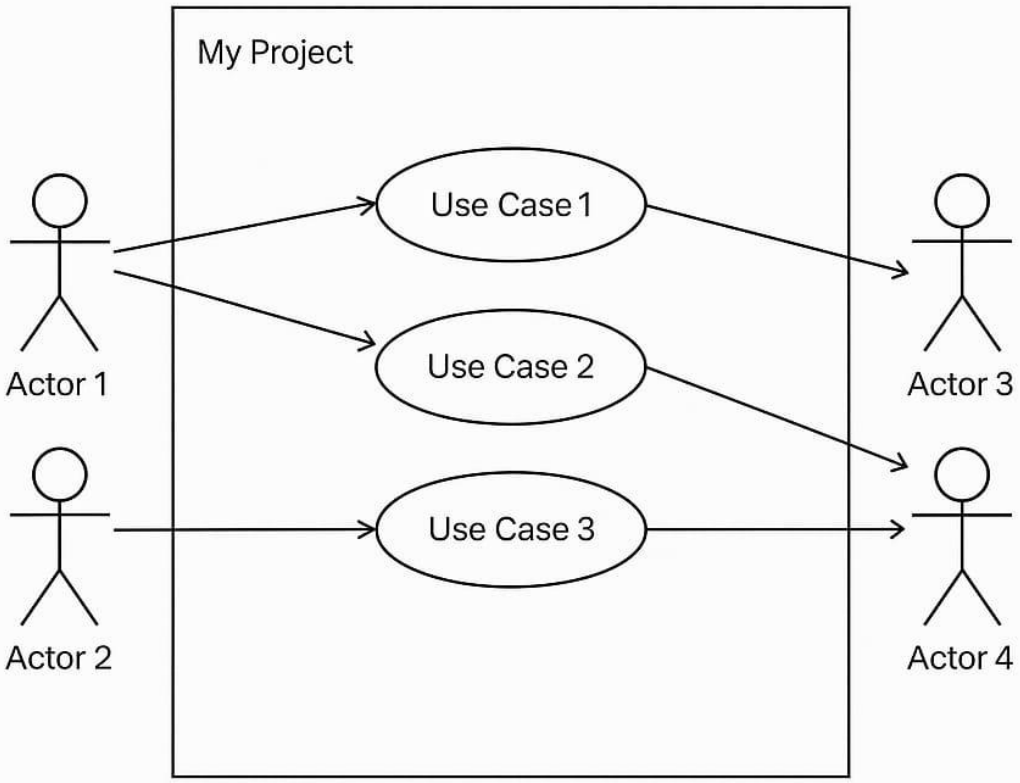


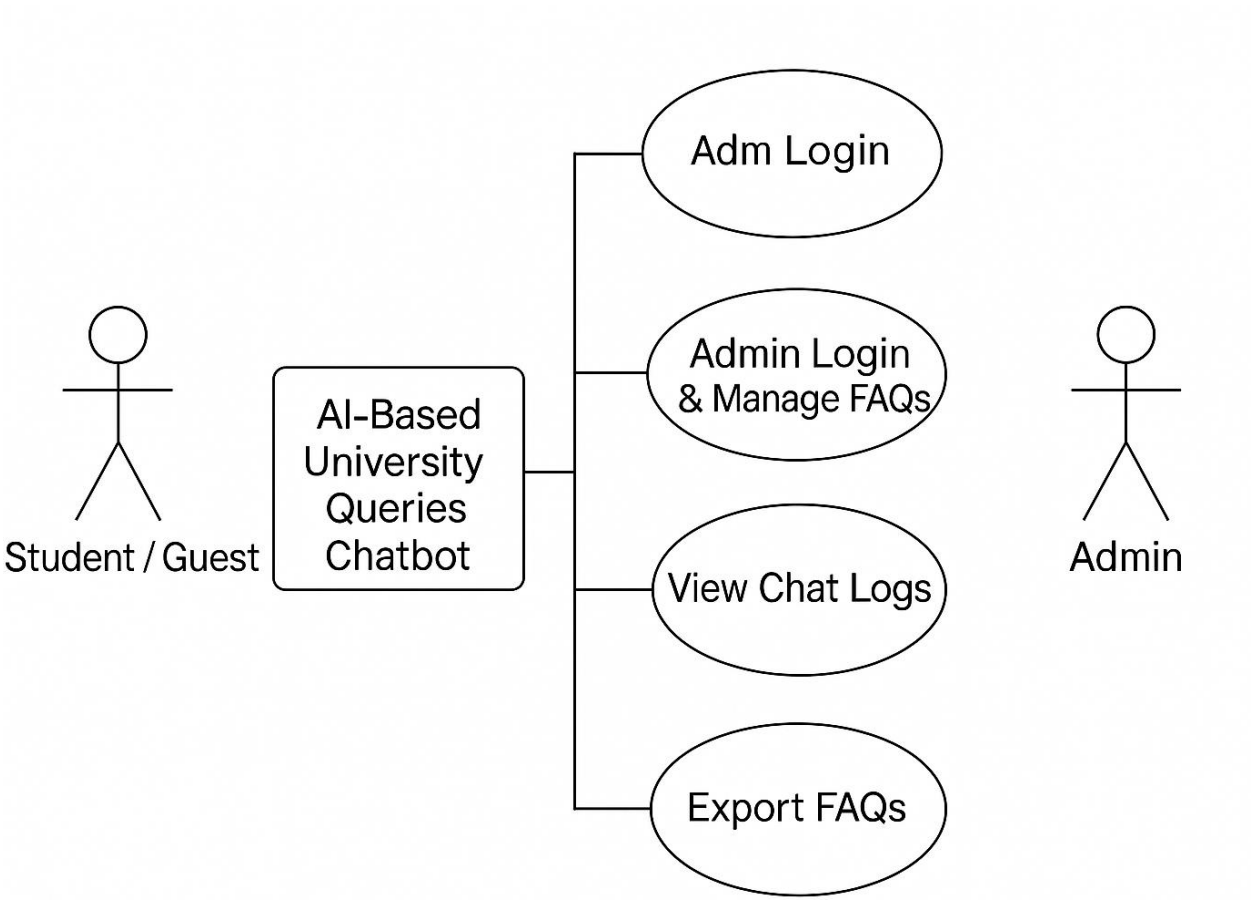
Figure 1: Use Case Diagram

3.2 use case 2

ID	Use Case Name	Actor(s)	Description
UC001	Admin Login & Manage FAQs	Admin	The admin logs into the system and manages FAQs (add/edit/delete).
UC002	Ask Question	Student / Guest	The user asks a question and the chatbot provides an appropriate response.

UC003	View Chat Logs	Admin	The admin views the history of user-chatbot interactions.
UC004	Export FAQs	Admin	The admin exports the existing FAQs into CSV or PDF format.

Diagram :



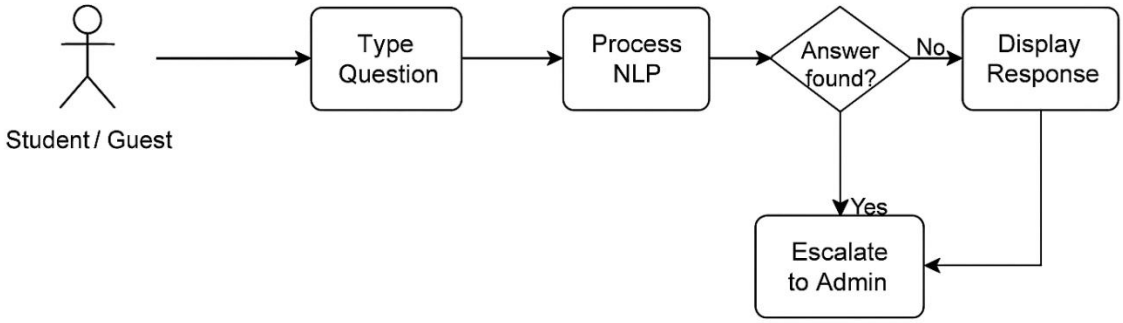
3.3 use case 3

Activity Diagram Description:

This activity diagram represents the workflow for the 'Ask a Question' feature in the system. The process begins when a user, either a student or a guest, submits a question by typing it into the system interface. Once the question is entered, it undergoes Natural Language Processing (NLP) to analyze and understand the intent and context of the query.

After processing, the system attempts to find a relevant answer from its knowledge base or database. A decision is then made based on whether an answer is found:

- **If an answer is found**, the query is escalated to an administrator for further review or action, ensuring the response is accurate and appropriate.



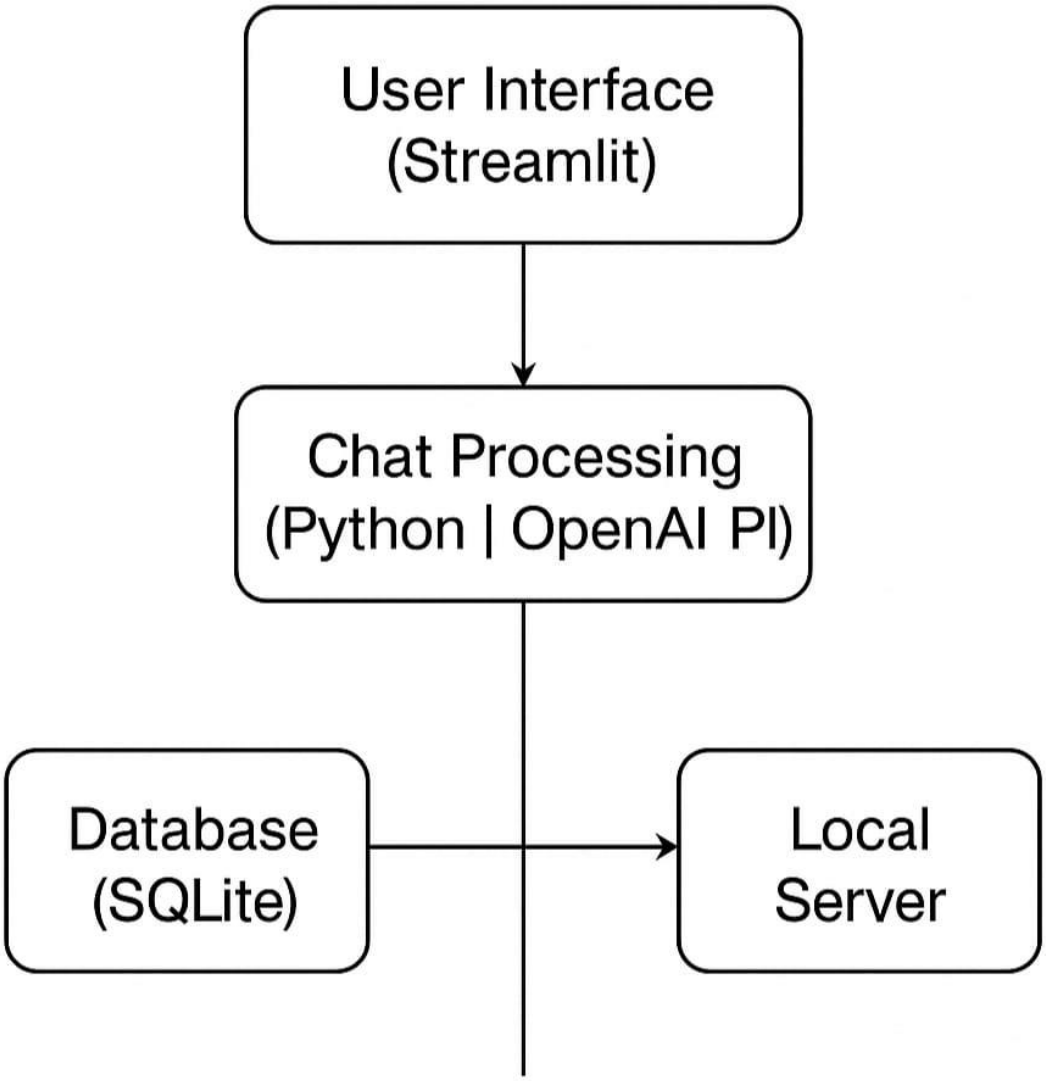
Activity Diagram for 'Ask a Question'

4. References

1. OpenAI API Documentation: https://platform.openai.com/docs
2. Streamlit Documentation: https://docs.streamlit.io
3. SQLite Documentation: https://sqlite.org/docs.html

5. Appendix: System Architecture Diagram

The following diagram illustrates the high-level architecture of the AI-based University Queries Chatbot. It highlights the interaction between user interface, processing layer, OpenAI integration, and the local database.



AI System Architecture Diagram